GLM Basics Guide

Overview

GLM (General Lake Model) is a 1D hydrodynamic model that simulates vertical temperature and mixing dynamics i

Running GLM

```
cd /root
glm
```

GLM reads glm3.nml in the current directory and produces output in output/output.nc.

Input File Structure

glm3.nml - Main configuration file (Fortran namelist format)
bcs/*.csv - Boundary condition files (meteorology, inflows, outflows)

Configuration File Format

glm3.nml uses Fortran namelist format with multiple sections including glm_setup, light (Kw), mixing (coef_mix

Modifying Parameters with Python

Use a small regex-based helper to replace numeric values for named parameters in glm3.nml, then rewrite the fi

Common Issues

GLM fails to start: missing input files; check bcs/ directory.
No output generated: invalid nml syntax; check namelist format.
Simulation crashes: unrealistic parameters; use values within valid ranges.

Best Practices

- Always backup glm3.nml before modifying.
- Run GLM after each parameter change to verify it works.
- Check output/ directory for results after each run.